

Table 1. Self-recognition AUROC scores (%) comparing our activation-based classifier (layer index in parentheses) against a perplexity baseline on the XL-Sum dataset. Both methods achieve near-perfect performance when conditioned on the original prompt, but only our method generalizes to the prompt-agnostic setting. Higher is better.

Model	With prompt		No prompt	
	Ours	PPL	Ours	PPL
Ministral-3-8B	100	99.71	99.99	32.33
Llama-3.1-8B	99.99	99.19	99.16	47.86
Llama-3.2-3B	99.96	99.43	99.03	47.49
Llama-3.2-1B	99.82	97.07	98.58	52.27

than in conventional free-form text-generation tasks, where AI-GTD is typically evaluated.

In this experiment, we evaluate self-recognition using 8 192 news articles and reference summaries from the English subset of the XL-Sum dataset (Hasan et al., 2021). We consider only articles at most 2 048 characters long to reduce arbitrariness in the summaries and to allow LLMs to produce high-quality summaries for each article.

As a baseline, we use each LLM’s perplexity (Jelinek et al., 2005) as an intrinsic measure of its predictive performance. For a token sequence $x_{1:T}$, we compute

$$\text{PPL}(x) = \exp\left(-\frac{1}{T} \sum_{t=1}^T \log p_M(x_t \mid x_{<t})\right),$$

where p_M denotes the conditional probability distribution defined by model M . Lower perplexity indicates that the model assigns higher likelihood to the text. We perform authorship attribution by scoring each summary with its perplexity and sweeping a decision threshold over the full range of scores to compute the AUROC, expecting the generating model to assign lower perplexity to its own summaries than to human-authored ones. While perplexity is commonly used in AI-GTD, it is a limited signal and is often augmented with additional statistics or computed in a contrastive fashion. For example, Hans et al. (2024) propose a zero-shot detector that contrasts two closely related LLMs by combining perplexity with a *cross-perplexity* score, which helps calibrate prompt- and topic-induced variation. We nevertheless report perplexity because it is computable from a single LLM without auxiliary models or features, making it a natural baseline for self-recognition.

Table 1 presents self-recognition capabilities across LLMs in this scenario. Our findings indicate that internal activations exhibit a strong, linearly separable signal for distinguishing LLM-generated from human-authored summaries, even for very short completions. Classifiers trained on these activations consistently outperform the perplexity-based

Table 2. F1 scores of the attribution task on ELI5 and Fresh News datasets across multiple LLMs. Results are averaged over five runs with different random seeds and steering vectors. The task measures the ability to classify texts generated by two independently steered model variants. Higher is better.

Model	ELI5		Fresh News	
	Token	Text	Token	Text
Ministral-3-8B	99.3	100	94.6	100
Llama-3.1-8B	94.0	99.1	90.5	99.1
Llama-3.2-3B	83.3	95.5	75.5	88.3
Llama-3.2-1B	72.0	85.3	69.5	83.8

baseline, though the baseline performs only marginally worse when prompts are available. Crucially, the activation-based signal remains robust in the prompt-agnostic setting, with AUROC decreasing by at most one point. By contrast, for Llama-family models, perplexity-based baselines do not consistently perform above chance without prompt access. Notably, for Ministral-3-8B-Instruct in this setting, the perplexity ordering is largely inverted, as human-authored summaries tend to attain *lower* perplexity than the model’s own generations. This behavior may reflect data overlap between the evaluation corpus and the pre-training or instruction-tuning data used for Ministral-3-8B-Instruct.

Additional experiments described in Appendix E indicate that layer-wise LDA performance remains largely consistent beyond the first layer, suggesting that self-recognition signals are distributed throughout the network. Further, classifiers trained on specific news domains generalize well to others, indicating that the learned signal is not strongly confounded by semantics.

3.2. Distinguishing Multiple LLMs with Steering

To validate the feasibility of the activation-based watermark described in Section 2.4.2, we conduct a controlled attribution experiment. We steer two identical model instances using two distinct random sparse vectors to generate completions for a fixed set of 1 000 prompts, with a maximum sequence length of 512 tokens. A classifier is then trained on the collected activation data to distinguish the source identity of the generated text.

To isolate the intrinsic detectability of the steering signal from extensive hyperparameter optimization, we fix the amplification coefficient to $\alpha = 5$ and the vector sparsity to 99.7% across all model families. While these parameters may not be optimal for every specific architecture, fixing them enables a direct comparative assessment of signal recovery across scales. Table 2 presents the attribution accuracy, revealing a clear trend: detection perfor-

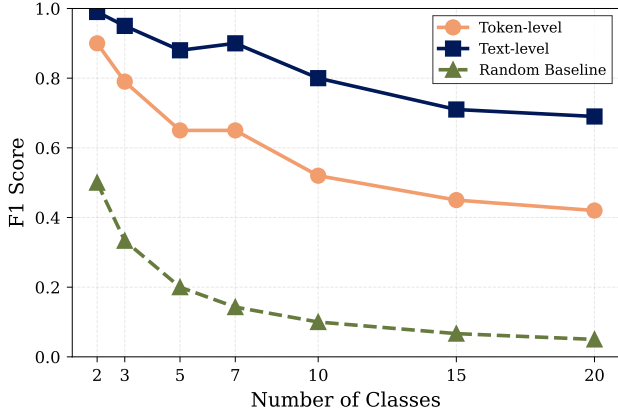


Figure 2. Performance of our attribution method across different numbers of classes. We compare text-level attribution with majority voting, token-level attribution, and a random-attribution baseline. F1 scores are shown for Llama-3.1-8B on our Fresh News dataset.

mance improves with model size. Interestingly, the question-answering dataset ELI5 yields higher separability than open-ended text generation tasks. This finding is counterintuitive, as the lower entropy of factual generation theoretically offers a more constrained manifold for embedding a watermark. This performance gap could therefore stem from the instruction fine-tuning of the underlying models.

We further evaluate the method’s scalability by increasing the pool of distinct steered identities. As shown in Figure 2, while the discrimination task becomes harder as the number of classes grows, performance remains significantly above the random baseline. While the current classification over N vectors shows limitations in scaling, these results imply the potential for a combinatorial multi-bit watermark, where multiple sparse vectors are superimposed for steering and detected independently. This would theoretically allow the encoding of 2^N identities.

Regarding generation quality, perplexity in this setting acts primarily as a measure of distributional shift from the base model rather than a direct proxy for text quality. To ensure that steering does not degrade generation quality, we calibrate the steering parameters using an external evaluator. We employ the quality-classifier-deberta model, a fine-tuned DeBERTa-v3 (He et al., 2023) from the NVIDIA NeMo Curator toolkit. In addition, we evaluate the steered models on the MMLU benchmark (Hendrycks et al., 2020) to verify minimal performance degradation. A comprehensive analysis of the quality trade-off is provided in Appendix F.

3.3. Robustness Study

AI-GTD often lacks robustness to modifications of the generated text, which can erase the watermark or the

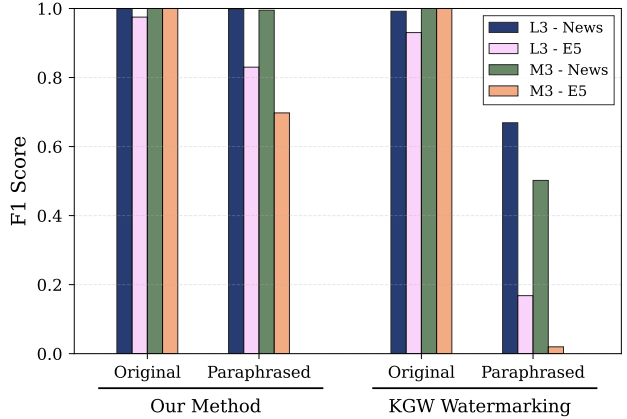


Figure 3. F1 scores for AI-GTD, comparing our method with traditional watermarking on the original LLM-generated text and a paraphrased version. Scores are computed with Llama-3.1-8B (L3) and Ministral-3-8B (M3) on the Fresh News (News) and ELI5 (E5) datasets.

LLM’s fingerprint. Automatic paraphrasing tools, such as DIPPER-XXL (Krishna et al., 2023), have been designed to evaluate this robustness and are used here. DIPPER-XXL reformulates text at the level of multiple sentences, rather than relying only on word substitution or sentence-level rewriting. Figure 3 reports F1 scores for distinguishing human-written from watermarked text, with and without paraphrasing, and compares our method to the green-list approach of the KGW watermark (Kirchenbauer et al., 2023). Our method shows greater robustness to paraphrasing, particularly in the free-form generation setting, where performance remains largely stable. One possible explanation is that our approach operates in a higher-level representation space, where attribution signals are aggregated across many tokens, potentially leading to more robust detection.

Here, we paraphrase only the LLM-generated text to wash out the watermark or steering signal. If we also paraphrase the human text, the F1 score of our method is slightly reduced, as paraphrasing human text acts as a spoofing mechanism for our detector. More details are given in Appendix C.

3.4. Influence of Steering-Vector Sparsity

Excessive steering intervention introduces significant noise into the generation process, often manifesting as repetitive loops or incoherent text. While these distinct artifacts make the attribution task trivial, they render the watermark unusable for practical applications where text quality is essential. Consequently, the optimal steering configuration must maximize detectability while minimizing quality loss.

To demonstrate the empirical advantages of sparse steering vectors over dense ones, we evaluate the trade-off they offer between attribution performance and generation qual-

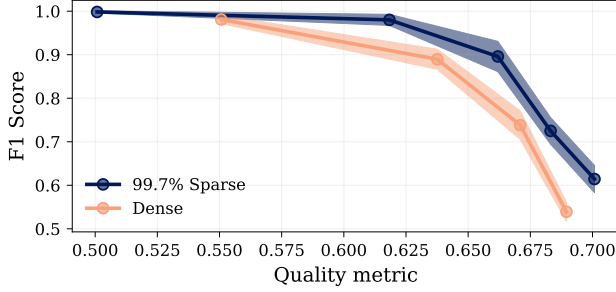


Figure 4. Trade-off between generated text quality and accuracy in distinguishing vanilla from steered generations of Llama-3-8B. The figure compares the effect of a 99.7% sparse steering vector with a dense counterpart. Points are obtained by varying the steering coefficient and are averaged over five random seeds, with all other settings held fixed.

ity. Since the multi-model attribution task described in Table 2 yields saturated accuracy for large models, we adopt a more challenging binary classification setting to better differentiate the steering types: distinguishing the unsteered Llama-3-8B base model from a steered counterpart. We perform a sweep over the amplification factor α , averaging the performance across five distinct steering vectors per configuration. The quality metric is computed using the quality-classifier-deberta model. The results in Figure 4 indicate that highly sparse vectors achieve a favorable trade-off profile compared to their dense counterparts, despite modifying only a fraction of the dimensions. This suggests that broad interventions are unnecessary, highlighting that targeted activation engineering offers a more robust approach to model steering.

3.5. Direct Signal Recovery via Cosine Similarity

Unlike standard activation engineering, which identifies specific directions to inhibit or amplify behavioral traits, such as sentiment or hallucination, our method operates on a fundamentally different principle. We inject a random, sparse steering vector. In the high-dimensional geometry of LLM activations, random vectors are, with high probability, approximately orthogonal to the semantic manifolds governing standard model behaviors, a property of concentration of measure (Vershynin, 2026). Therefore, we hypothesize that this orthogonality allows the steering signal to coexist with semantic content without significant interference.

Remarkably, this steering signal persists through the discretization step of token sampling and re-embedding. When the generated text is fed back into a non-steered base model, the steering vector can be directly retrieved from the resulting activations, without requiring a trained classifier. We observe a significant cosine similarity between the original steering vector and the activations of the base model processing the watermarked text. This implies that the trans-

Table 3. Accuracy for source attribution at the token and text levels, with and without paraphrasing. We compare simple cosine similarity attribution with the trained MLP approach under identical evaluation conditions. Higher is better. The best score for each method is underlined.

Detection Method	Granularity	Original Text	Paraphrased Text
Similarity	Token-level	62.9	58.6
Similarity	Text-level	<u>84.6</u>	77.8
MLP	Token-level	90.5	80.0
MLP	Text-level	<u>99.1</u>	89.3

formation from activation space to discrete tokens and back to activation space preserves the directional alignment of the injected signal. As illustrated in Figure 5, this alignment remains robust even after strong paraphrasing via DIPPER-XXL. The figure reports the average cosine similarity across all tokens in sequences up to 512 tokens.

This persistence enables a training-free detection mechanism. Instead of training a supervised MLP, we can attribute a text to a specific steered model by computing the cosine similarity between the collected activations and candidate steering vectors. This yields an attribution score for each token and, via majority voting, a global attribution decision. Table 3 presents the accuracy of this zero-shot approach for distinguishing between two random vectors. While it does not reach the near-perfect performance of the trained MLP, the accuracy is surprisingly high for a simple geometric metric. This confirms that the sparse signal is not merely a statistical artifact but is physically encoded into the generated sequence.

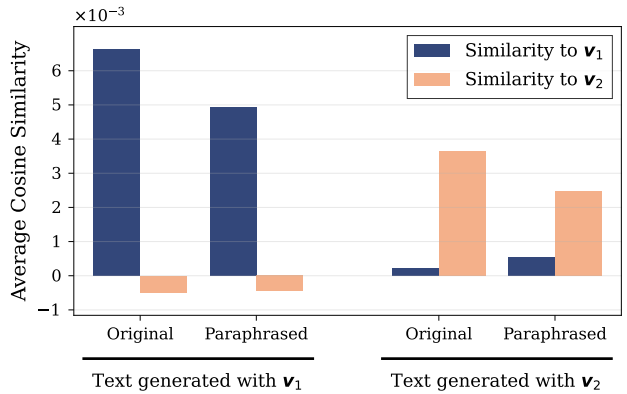


Figure 5. Example of alignment between steering directions and representations of induced text. Averaged cosine similarity between a pair of randomly sampled steering vectors (v_1 and v_2) and activation representations extracted from text generated by the model when steered along these directions.